

FAN, Sixing

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Objective Statement

To obtain an engineering position in the artificial intelligence industry that enables me to utilize my engineering degree, communication skills, and solid mathematical reasoning.

Education

CUHK, BSc in Mathematics and Information Engineering Sep 2023 – Present

- S.H.Ho College Outstanding Student Scholarship 2025/2026
- S.H.Ho College Master's List Award 2025
- Faculty of Engineering Dean's List 2024-2025

Projects

Gradient-Based Adversarial Attacks: FGM, FGSM and PGD Nov 2025

- Investigated three gradient-based adversarial attack methods through mathematical formulation
- Conducted empirical evaluation by Robust Bench framework on ImageNet-1k validation set using ResNet-18
- Evaluated model robustness using transfer accuracy and false confidence metrics under varying perturbation bounds (2/255, 4/255, 8/255)
- Demonstrated that PGD exhibits the strongest attack capability

Robot Navigation, Research Assistant, CUHK Information Engineering Department Jun – Aug 2025

- Formulated Static Scene Complexity Score (SSCS) to evaluate indoor scene complexity for robot navigation
- Developed SSCS optimizer using Python for generating indoor scenes with varying complexity levels
- Implemented interpolation sampling and Bayesian optimization to improve system efficiency by 30%
- Conducted navigation experiments in simulated environments using scenes generated by the SSCS optimizer

Internship Experience

IT Support, Tech Vision System Limited, Hong Kong Jun – Aug 2024

- Completed Huawei IdeaPresence project according to client specifications and timelines
- Configured Huawei IdeaHub products (S2, ES2, ES2 Plus) for enterprise clients
- Managed company website content updates and social media operations
- Provided technical support and troubleshooting for IT infrastructure

Skills

Languages: Mandarin (Native), English (Fluent), Cantonese (Fluent)

Tools: VS Code, Cursor, GitHub, Blender, Linux System and Wireshark/tcpdump

Programming: Python, Java, C, MATLAB, LaTeX

Frameworks: NumPy, Matplotlib, PyTorch, bpy, Scikit-learn

Mathematics: Calculus, Linear Algebra, Complex Analysis, Discrete Mathematics and Probability

Extracurricular Activities

Fitness and Mentorship Scheme, CUHK Jan – Apr 2024

Student Volunteer

- Learned and played dodgeball and cricket with youth from ethnic minority groups
- Organized monthly sports activities and mentorship sessions for 10+ participants
- Facilitated cultural exchange and communication to support minority youth integration into the community